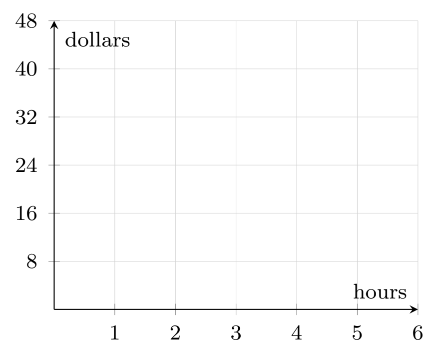
Graphing Proportional Relationships

Plotting a proportional relationship from a table, finding the unit rate from a graph, and writing the equation of a line through the origin

Before you start.

- A relationship is **proportional** when its graph is a straight line *through the origin* $(0, 0)$.
- The **unit rate** (constant of proportionality) k is the rise over the run measured from the origin: $k = \dfrac{y}{x}$ for any point on the line except $(0, 0)$.
- The equation is always $y = kx$. State the units of k whenever the problem gives them.

1. Dev is paid the same amount for every hour he walks dogs.

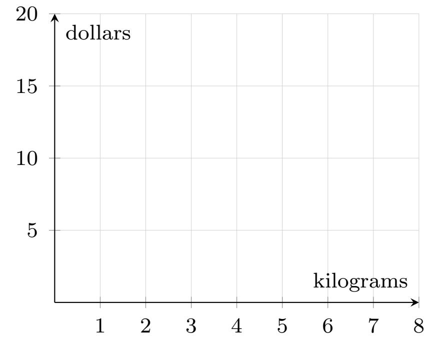


Hours worked	0	3	5
Dollars earned	0	24	40

Plot the three points and draw the straight line through them. Then find the unit rate, in dollars per hour.

Answer: ________________

2. Rice is sold loose, so the cost is proportional to the mass.

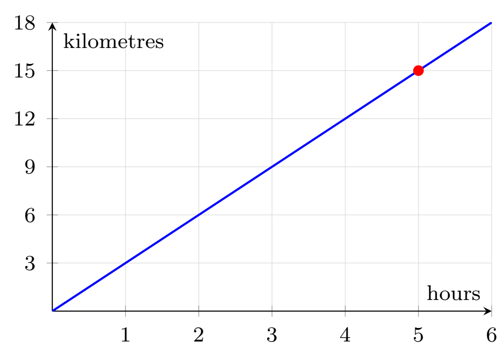


Kilograms	0	2	6
Cost in dollars	0	5	15

Plot the three points and draw the line through them. Then find the cost of one kilogram of rice, in dollars.

Answer: ____________________

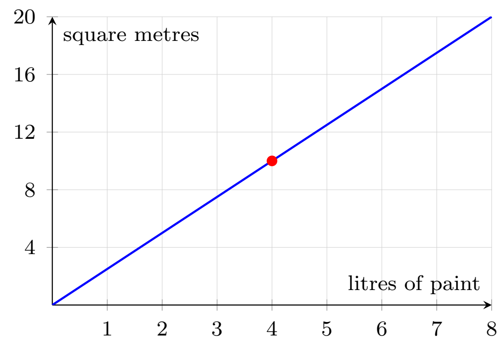
3. The graph shows how far a cyclist travels over time. The line passes exactly through the marked point.



What is the cyclist's unit rate, in kilometres per hour?

Answer: ____________________

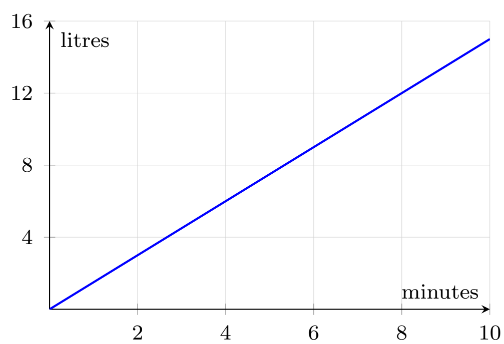
4. The graph shows the amount of paint needed to cover a wall. The line passes through the marked point.



Write the equation of this line in the form $y = kx$, where x is litres of paint and y is square metres covered.

Answer: ________________

5. A tap fills a barrel at a constant rate. The graph of the relationship is drawn below.

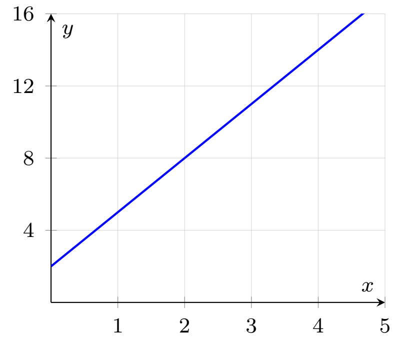


The equation of this line is $y = 1.5x$. Use the graph to check your answer, then solve to find the number of minutes it takes to collect 12 litres.

Answer: ________________

6. Not every straight line is proportional. This table and its graph belong together.

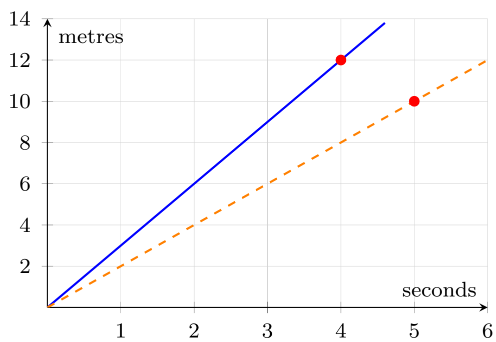
x	1	2	4
y	5	8	14



Work out $y \div x$ for each column. Are the three quotients equal? Say whether the relationship is proportional, and name the feature of the graph that settles it.

Answer: ________________

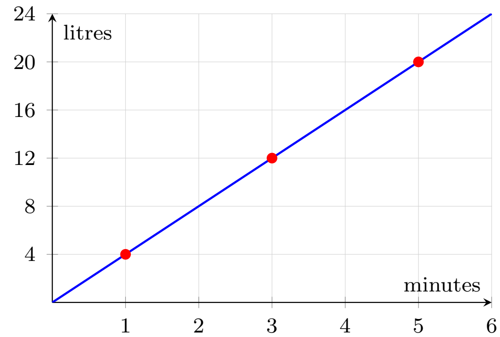
7. Two runners keep a steady speed for the whole race. Runner A is the steeper line; runner B is the shallower one.



Each line passes through a marked point. Find both unit rates, in metres per second, and say which runner is faster.

Answer: ________________

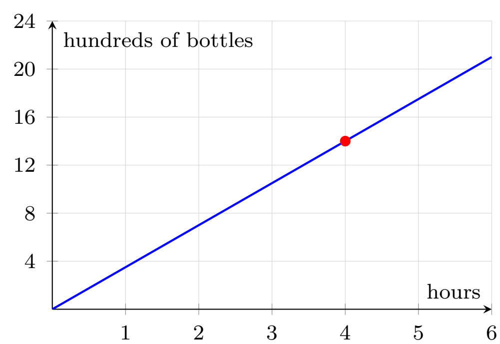
8. Water runs into an empty tank at a constant rate.



Read the graph at 1 minute and at 5 minutes. How many more litres are in the tank at 5 minutes than at 1 minute, and what is the unit rate in litres per minute?

Answer: ________________

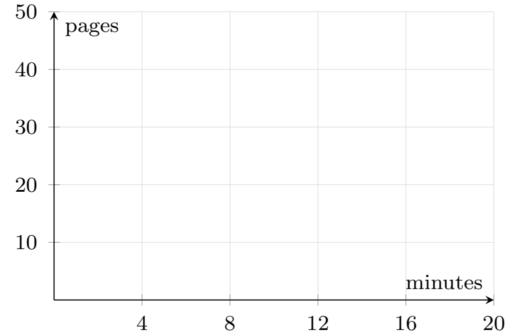
9. A machine bottles juice at a constant rate. The graph shows the first few hours only.



Find the unit rate from the marked point, then use the equation to predict the output after 12 hours. The prediction runs past the end of the drawn axis, so the graph alone cannot answer it.

Answer: ________________

10. A printer prints 5 pages every 2 minutes, and it starts from a printed count of zero.



Draw the graph of this relationship on the grid, starting at the origin. Then use its equation to find how many minutes it takes to print 45 pages.

Answer: ________________